

Anomaly_Detection_in_El...
PDF

Can you review the following scholarly article? I have some questions on the machine learning methods used and the features they used with the various modeling techniques.

Yes — I've reviewed the article you uploaded:

 Anomaly_Detection_in_Electronic...

I'll give you a **structured, technically grounded overview** focused specifically on:

1. Machine learning methods used
2. What features/data they actually modeled
3. How those features interact with each method
4. Where the design is strong vs. questionable (important for your OHDSI context)

1. What ML methods were used (and why)

The paper uses a **5-model ensemble** ("Voting Machine"):

- **Local Outlier Factor (LOF)** → density-based
- **Isolation Forest (IF)** → tree-based isolation
- **Gaussian model** → probabilistic (distribution-based)
- **KNN distance-based**
- **Prophet** → time-series forecasting

These are combined via **majority voting (threshold C)**  Anomaly_Detection_in_Electronic...

👉 Key design principle:

Each model captures a *different notion of "anomaly"* (density, isolation, deviation, distance, temporal trend)

2. What features are actually used

This is *critical* — and somewhat subtle.

Core feature: univariate time-series volume

They do **NOT** model patient-level features.

Instead, they aggregate:

- Count of EHR "orders" per:
 - **facility (station)**
 - **service type** (e.g., nursing, surgery, pharmacy)
 - **time (daily)**

👉 So each model operates on:

$V = \{v_1, v_2, \dots, v_n\}$



Where each v_i is a **daily count**  Anomaly_Detection_in_Electronic...

Derived features (implicit, not explicit feature engineering)

They don't build feature vectors — but they *implicitly construct features* via:

1. Temporal context (sliding window)

- Applied to non-temporal models (LOF, IF, KNN, Gaussian)
- Converts time series into **local windows**

👉 This is effectively:

- lag features
- local distribution context

2. Normalization

- **Min-max scaling** across series  Anomaly_Detection_in_Electronic...

👉 This standardizes:

- across hospitals
- across service types

3. Lookback mean (for graph stage)

Used later in graph weighting:

$$|o_i - \text{mean}_{\text{lookback}}|$$

👉 This becomes a **deviation feature** for system-level modeling

4. Temporal alignment (MaxDeferTime)

- Defines whether anomalies are “simultaneous”

👉 This introduces a **temporal correlation feature across nodes**

Key takeaway:

The **feature space is extremely simple**:

- ◆ Univariate time-series + local temporal context
- ◆ No multivariate clinical features
- ◆ No patient-level modeling

3. How each model uses those features

This is where things get interesting.

◆ LOF (density-based)

- Uses **local density vs neighbors**
- Input: windowed values

👉 Detects:

- local dips/spikes relative to neighbors

⚠️ Weakness:

- struggles with global shifts (e.g., system-wide changes)

◆ Isolation Forest

- Random splits isolate rare points quickly

👉 Detects:

- sparse, extreme values

⚠️ Weakness:

- sensitive to scaling + distribution shape
-

◆ Gaussian model

- Assumes:

$$x \sim \mathcal{N}(\mu, \sigma)$$

👉 Detects:

- deviations from expected distribution

⚠️ Major limitation:

- ignores temporal order  Anomaly_Detection_in_Electronic...
-

◆ KNN distance

- Uses distance to k-th neighbor

👉 Detects:

- points in sparse regions

⚠️ Weakness:

- high-dimensional curse (less relevant here since univariate)
-

◆ Prophet (time-series model)

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

👉 Explicitly models:

- trend
- seasonality
- residuals

👉 Detects:

- deviations from expected temporal pattern

⚠️ Limitation:

- smooths abrupt changes (may miss spikes)
-

🧩 4. Ensemble (Voting Machine)

They define:

$$y = 1 \text{ if } \sum_j P_j > C$$

- Each model outputs binary anomaly
- Threshold **C controls sensitivity**

👉 Behavior:

- **Low C** → high recall, more false positives
 - **High C** → high precision, fewer detections
-

Why this works well

From results:

- Ensemble outperforms all models individually  Anomaly_Detection_in_Electronic...

Reason:

- Models are **weakly correlated detectors**
 - Voting reduces:
 - model-specific bias
 - false positives
-

5. Graph-based features (system-level modeling)

This is the most novel part.

They convert anomalies → graph:

Nodes:

- hospitals

Edges:

- similarity of **co-occurring anomalies**

Edge weight:

$$\text{SAS}(o_i, o_j) = \log(|o_i - \mu_i| + 1) \cdot \log(|o_j - \mu_j| + 1)$$

 Features used here:

- magnitude of deviation
 - temporal proximity
 - shared service type
-

Then:

- threshold graph (spectral method)
- detect clusters (paralclique)

 Output:

- **system-level anomaly clusters**
-

6. Critical evaluation (important for your work)

Strengths

1. Simple but scalable feature design

- Works across 130 hospitals, millions of records
- No need for feature engineering

2. Ensemble reduces false positives

- Very relevant to data quality pipelines

3. Graph layer = key innovation

- Moves from:
 - **point anomalies → system anomalies**

 This aligns strongly with:

- OHDSI network studies
- distributed data quality signals

 Weaknesses (important)

1. No clinical semantics

- Only volume counts
- No:
 - measurement values
 - concept-level features
 - patient-level signals

 For your **measurement domain work**, this is a major gap

2. Univariate modeling

- Ignores relationships between:
 - service types
 - hospitals
 - patient cohorts

3. Weak temporal modeling (except Prophet)

- Sliding window is a workaround
- Not true temporal modeling

4. Feature engineering is minimal

- No:
 - seasonality decomposition (outside Prophet)
 - autocorrelation features
 - domain-driven signals

5. Ensemble is heuristic

- Equal weights for all models
- No learned weighting

 7. Mapping to your OHDSI / measurement use case

This is where it gets interesting for you.

This paper:

- Detects **system-level anomalies in activity volume**

Your work:

- Detects **data quality anomalies in measurement values**

 Key difference:

Paper	Your work
Volume counts	Measurement distributions
Operational anomalies	Clinical plausibility
Time-series	Concept-level distributions

Paper	Your work
Graph across hospitals	Potentially graph across concepts

 Big opportunity for you

You could extend this by:

1. Replace volume features with:

- measurement distributions
- value ranges
- percentiles

2. Add multivariate features:

- concept relationships
- temporal drift

3. Replace voting with:

- weighted ensemble
- meta-model

4. Extend graph layer:

- concept co-anomalies
- site-level + concept-level hybrid graph

 Bottom line

- The ML methods are **standard but complementary**
- The **feature space is intentionally simple** (univariate time-series counts)
- The **real novelty is the graph layer**, not the ML models
- The approach is **strong for system monitoring**, but **limited for clinical data quality**

 If you want to go deeper

You mentioned you have questions — feel free to ask things like:

- “How would LOF behave on measurement values vs volumes?”
- “How would you redesign the feature space for OMOP measurement?”
- “Which of these models are actually appropriate for OHDSI DQD extension?”

Happy to go deep — this is very aligned with your anomaly detection + OHDSI work.

Sources

     ...

I'm exploring the use of LOF for my data quality problem for measurement values. I'd like to explore the feature space of the OMOP CDM. To keep things simple, I have a data set that uses features such as age, gender and care site. However, it doesn't seem to be very useful. I was hoping to better understand the research I shared to see how I might translate this into my problem space. Any suggestions?

You're asking exactly the right question—and your instinct is correct:

LOF + (age, gender, care_site) is almost guaranteed to underperform for measurement data quality.

Let me connect what the paper is *actually doing* to how you can rethink your feature space in OMOP.

Why your current feature space isn't working

LOF works by comparing **local density in feature space**.

So the key question is:

👉 What defines "neighborhood similarity" for your problem?

Right now:

- Age → weak signal unless strongly stratified
- Gender → binary, low information
- Care site → often noisy / heterogeneous

👉 These **don't define a meaningful "clinical neighborhood"** for measurement values.

So LOF is essentially operating on **uninformative geometry**.

What the paper does differently (and why it works)

The paper avoids this problem entirely by:

! **Not using demographic features at all**

Instead, they define similarity using:

✅ **Behavior-based features:**

- Time-series patterns
- Deviations from expected values
- Temporal context (sliding windows)
- Magnitude relative to historical baseline

👉 Their "neighborhood" is:

"How does this behave relative to its expected pattern?"

—not who the patient is.

Translate that idea to OMOP measurement data

Instead of:

(age, gender, care_site)



You want:

🔥 **features that define expected measurement behavior**

Better feature space for LOF (measurement-focused)

Think in 3 layers:

1. Value-centric features (MOST important)

These define plausibility.

For each measurement:

- Raw value (normalized)
- Z-score within concept:

$$z = \frac{x - \mu_{concept}}{\sigma_{concept}}$$

- Percentile within concept
- Distance from reference range (if available)
- Log(value) for skewed labs

👉 This gives LOF a **clinical density structure**

🕒 2. Temporal features (paper-inspired)

This is where you directly borrow from the paper.

- Change from prior value:

$$\Delta x = x_t - x_{t-1}$$

- Rolling mean / median deviation
- Slope over last N observations
- Volatility (std over window)

👉 Equivalent to their:

- sliding window
- deviation from lookback mean  Anomaly_Detection_in_Electronic...

🧪 3. Contextual clinical features

Instead of demographics, use:

- Measurement concept ID (encoded)
- Unit consistency flag
- Measurement type (lab vs vital)
- Visit type (inpatient/outpatient)
- Condition context (optional)

👉 These define **clinical context**, not just population grouping

🚨 Key insight (this is the big one)

! LOF needs **feature space where anomalies are sparse relative to similar observations**

Your current setup:

- Defines similarity by *people*

Better setup:

- Defines similarity by *measurement behavior*

🔬 Example: Creatinine

Instead of:

(age=65, male, site=A)



Use:

```
value = 2.1
z_score = +3.2
delta = +0.8 from prior
rolling_mean_diff = +1.5
unit_flag = consistent
```

- 👉 Now LOF can detect:
- sudden spikes
 - implausible values
 - inconsistent trajectories

 Mapping back to the paper

Paper	Your adaptation
Time-series volume	Measurement trajectories
Lookback mean	Patient or cohort baseline
Sliding window	Rolling features
Voting ensemble	(optional) multiple detectors
Graph across hospitals	Graph across concepts or sites

 Should you even use LOF?

Short answer: **Yes—but only after fixing feature space**

LOF is best for:

- local anomalies
- subtle deviations
- dense data regions

LOF struggles with:

- global shifts (e.g., unit errors across site)
- extreme outliers (Isolation Forest better)

👉 Which leads to...

 Strong recommendation (inspired by paper)

Don't use LOF alone.

Use a mini "Voting Machine" like they did:

- LOF → local anomalies
- Isolation Forest → global anomalies
- Simple Gaussian → distribution anomalies
- Temporal model → trend anomalies

👉 This directly mirrors their architecture  Anomaly_Detection_in_Electronic...

 Extending their graph idea to OMOP (high value for you)

This is where your work can go beyond the paper.

Instead of:

- graph of hospitals

Build:

- graph of **concepts or sites**

Edges = co-occurring anomalies:

- same concept across sites
- same site across concepts

👉 This helps detect:

- unit mapping issues
- ETL problems
- vocabulary drift

⚠️ Common trap (you're currently hitting it)

Using **convenient features** instead of **mechanistically meaningful features**

Demographics are easy—but:

- they don't define measurement plausibility
- they don't create useful density structure

✅ Practical next step (concrete)

Try this minimal feature set:

```
features <- data.frame(
  value_scaled,
  z_score_concept,
  delta_prev,
  rolling_sd,
```

↔ R



Run LOF on that.

Then compare to:

```
features_demographic <- data.frame(
  age,
  gender,
```

↔ R



👉 You'll see a dramatic difference.

🌟 Bottom line

- The paper succeeds because it models **behavior**, not **entities**
- LOF only works if your feature space encodes **expected structure**
- For OMOP measurement:
 - **value + temporal + distribution features > demographics**
- The real opportunity is:
 - combining LOF with other detectors

- adding a graph layer for system-level anomalies

If you want, I can:

- Help you design a **feature engineering pipeline aligned to OMOP CDM**
- Or map this directly to **DataQualityDashboard-style checks + ML augmentation**

This is very close to what you’ve been exploring with anomaly detection + measurement plausibility.

Sources



Let's start from a different spot. I have a number of univariate outlier detection methods that I've applied to a single measurement concept. For that concept, I have the value of the measurement and the frequency of the measurement. I'd like to learn the plausible range of values for a given measurement using these methods: tukey fences, quantile thresholds, z-score, modified z-score and generalized esd. Could I use some type of voting approach or consensus model using these to figure out the plausible min/max in my data set? For this measurement concept, a normal range is well-defined but learning what is biologically plausible is my ultimate goal.

Short answer: **yes**—using a voting/consensus approach across univariate detectors is a **very reasonable way to learn a “plausible” range**, but you’ll want to structure it carefully so you don't just average biases from each method.

Let’s break this down in a way that maps cleanly to your OMOP measurement use case.

 1. What you’re *really* trying to estimate

You’re not just detecting outliers—you’re estimating:

A robust, data-driven plausible range (min/max) for a concept

This is subtly different from:

- statistical “normal range”
- or clinical reference range

 You want:

- **biologically plausible bounds**
- tolerant of real-world noise
- but resistant to ETL/unit errors

 2. Your candidate detectors (good selection)

You’ve chosen a strong mix:

Method	Strength	Weakness
Tukey fences	robust, distribution-free	symmetric, IQR-sensitive
Quantiles	simple, interpretable	arbitrary cutoffs
Z-score	good for Gaussian-ish	sensitive to skew

Method	Strength	Weakness
Modified Z-score (MAD)	robust to outliers	still assumes unimodal
Generalized ESD	formal statistical test	assumes approximate normality

👉 This is already an **ensemble of different assumptions**, which is exactly what you want.

🔥 3. Yes—use a voting / consensus approach (but do it right)

Inspired by the paper's **Voting Machine** [Anomaly_Detection_in_Electronic...](#), you can do:

Step 1: Each method produces bounds

For each method m :

$$[\min_m, \max_m]$$

Step 2: Convert to “flags” per observation

For each value x_i :

```
flag_m(i) = 1 if x_i is outside method m bounds
```



Step 3: Voting

$$flag(i) = \sum_m flag_m(i)$$

Then define:

- **strict**: flag ≥ 4 (out of 5)
- **moderate**: flag ≥ 3
- **lenient**: flag ≥ 2

🎯 4. Turning this into a plausible range

This is the key step (and where most people go wrong).

Instead of averaging bounds:

❌ Don't do:

```
plausible_min = mean(min_m)
```



👉 This mixes incompatible assumptions.

✅ Better approach: derive bounds from consensus labels

Step A: Identify consensus “valid” points

```
valid_points = {x_i where flag(i) < threshold}
```



Step B: Define plausible range from valid points

Then compute:

- `min_plausible = min(valid_points)`
- `max_plausible = max(valid_points)`

OR more robust:

- 0.1th percentile of `valid_points`
- 99.9th percentile of `valid_points`

🔥 This is your “learned plausible range”

👉 It reflects:

- agreement across methods
- robustness to distribution shape
- resistance to extreme noise

🧩 5. Incorporating frequency (this is important)

You mentioned you also have **frequency**.

This is a huge advantage.

Option A: weighted voting

Weight each value by frequency:

```
weighted_flag(i) = sum_m flag_m(i) * freq_i
```



👉 Rare extreme values won't dominate

Option B: density-aware filtering

Drop values where:

- frequency is extremely low
- AND flagged by multiple methods

👉 This removes:

- data entry errors
- unit mismatches

Option C (very powerful)

Define:

```
support(x) = frequency(x) / total
```



Then:

- keep values with:
 - high support
 - OR low support but not flagged

💡 6. Important nuance: plausible ≠ frequent

You need to explicitly allow for:

- rare but biologically plausible values

Example:

- potassium = 6.5 (rare but plausible)
- potassium = 50 (implausible)

👉 So your rule should be:

Remove values that are:

- flagged by consensus
- AND inconsistent with distribution structure

7. Suggested full workflow

Here's a clean pipeline:

Step 1: Compute bounds per method

```
bounds <- list(  
  tukey = c(min_t, max_t),  
  quantile = c(min_q, max_q),  
  z = c(min_z, max_z),  
  modz = c(min_mz, max_mz),
```

↔ R



Step 2: Flag values

↔ R



Step 3: Consensus score

↔ R



Step 4: Define valid set

↔ R



Step 5: Final plausible range

↔ R



🔥 8. Enhancements (this is where you go beyond the paper)

1. Asymmetric bounds

Most lab values are skewed.

👉 Compute lower/upper separately:

- lower: methods sensitive to low-end
 - upper: methods sensitive to high-end
-

2. Log-transform first

For skewed labs:

↔ R



Run detection → back-transform

3. Mixture-aware approach

If distribution is multi-modal:

- split first (e.g., inpatient vs outpatient)
 - or cluster values
-

4. Stability analysis

Bootstrap:

- recompute bounds on samples
 - check stability of min/max
-

⚠️ 9. Where this approach can fail

Be aware:

❗ If data is already corrupted

- consensus will learn wrong range

❗ If distribution is very skewed

- z-score / ESD distort results

❗ If units are mixed

- must normalize first (critical in OMOP)
-

🕒 10. Connection back to your broader work

This approach is basically:

- ◆ A univariate Voting Machine
 - ◆ Analogous to the paper's ensemble approach [Anomaly_Detection_in_Electronic...](#)
 - ◆ But adapted for **value plausibility** instead of **time-series anomalies**
-

✓ Bottom line

- Yes, a **consensus/voting approach** is absolutely appropriate
- But:
 - vote on *flags*, not *bounds*
 - derive range from *agreed valid data*
- Use frequency to:
 - suppress noise
 - preserve rare but real values

If you want, next step I'd suggest:

👉 I can help you turn this into:

- a reusable **HADES-style R function** (`learnPlausibleRange`)
- or integrate it into a **DQD-style check with ML augmentation**

This is very close to something publishable in the OHDSI space.

Sources



How can I share this chat with my Codex co-pilot?

Short answer: there's **no direct "send to Codex / Copilot" button yet**, so you have to pass the context manually. But there are a few *good workflows* depending on how you're using Codex.

🗨 Best ways to share this chat with Codex / Copilot

✓ Option 1 (fastest): Copy → paste into Codex

If you just want continuity:

1. Select this chat (Ctrl+A → Ctrl+C)
2. Paste into Codex prompt or a file
3. Add instruction like:

"Continue this discussion and help implement..."

👉 This is the most common workflow right now.

✓ Option 2 (recommended for coding): Save as Markdown + load into repo

This works *really well* with Codex.

Steps:

1. Copy chat → paste into a `.md` file
2. Save as:

plausible_range_design.md



3. Add to your repo
4. In Codex:

“Read plausible_range_design.md and help me implement this in R”



👉 Codex works best when context lives in files (not chat memory) Aakash News

✅ Option 3: Use ChatGPT “Share link”

1. Click Share (top right of chat)
2. Copy link
3. Paste into Codex

👉 Works if Codex can access external links

👉 But less reliable than local files OpenAI Help Ce...

✅ Option 4 (structured / reproducible): Export + reuse

You can export all chats:

- Settings → Data Controls → Export Data OpenAI Help Ce...
- Extract conversation
- Convert to .md or .txt
- Feed into Codex

👉 Best for long-term workflows or research reuse

🚫 What you *can't* do (yet)

- No native “sync ChatGPT → Codex memory”
- No automatic transfer of reasoning/history

This is actually a known limitation:

Codex does not currently have access to ChatGPT conversation memory GitHub

🔥 What I recommend for *your* use case

Given you're doing **method development** (plausible ranges, LOF, etc.):

👉 Use Option 2 (Markdown in repo)

Why:

- Keeps your thinking version-controlled
- Codex can reason over it repeatedly
- You can evolve it into:
 - documentation
 - methods section
 - R implementation

🧩 Pro tip (high leverage)

Before exporting, ask me:

“Summarize this conversation as a design spec for implementation”

Then:

- save that cleaner version
- feed into Codex

👉 You'll get much better results than dumping raw chat

✓ Bottom line

- Yes, you can share it—but manually
- Best approach:
 - copy → markdown → repo → Codex
- Treat the chat as:
 design context → not runtime memory

If you want, I can turn *this exact discussion* into a clean **HADES-style design doc + R function spec** that you can drop straight into Codex.

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Turn this chat into a markdown file I can share with Codex

Here is a **clean, structured Markdown file** you can drop directly into your repo and use with Codex. It's organized as a **design + implementation spec**, which will work much better than raw chat (and aligns with Markdown best practices: readable plain text, structured headings, fenced code blocks) Google GitHub +1

📄 Copy everything below into a file named:

plausible_range_consensus.md 

```
# Learning Plausible Measurement Ranges via Consensus Outlier Detection

## Context

This document summarizes a design discussion on how to learn biologically plausible value ranges

The goal is to move beyond:
- Clinical "normal ranges"
- Simple statistical thresholds

Toward:
> A robust, data-driven estimate of plausible min/max values

---

## Problem Statement

Given a dataset for a single measurement concept with:
- Measurement values

<> Markdown 
```

```
[min_m, max_m]

---

### Step 2: Flag observations
```

For each value `x\_i`:

flag_m(i) = 1 if $x_i < \min_m$ OR $x_i > \max_m$
 flag_m(i) = 0 otherwise

Step 3: Voting / consensus

Aggregate across methods:

score(i) = sum(flag_m(i))

Define thresholds:

- Strict: score ≥ 4
- Moderate: score ≥ 3 ← recommended
- Lenient: score ≥ 2

Step 4: Define valid data

valid_points = { x_i where score(i) < threshold }

Step 5: Learn plausible range

Preferred (robust):

min_plausible = quantile(valid_points, 0.001)
 max_plausible = quantile(valid_points, 0.999)

Alternative (less robust):

min_plausible = min(valid_points)
 max_plausible = max(valid_points)

Incorporating Frequency

Frequency provides important signal about reliability.

Option A: Weighted voting

weighted_score(i) = sum(flag_m(i)) * freq_i

Option B: Density filtering

Remove values that are:

- Low frequency
- AND flagged by multiple methods

Option C: Support-based filtering

$\text{support}(x) = \text{frequency}(x) / \text{total_observations}$

```

Keep values that:
- Have high support
- OR are not flagged

---

## Important Distinction

> Plausible ≠ Frequent

Examples:
- Potassium = 6.5 → rare but plausible
- Potassium = 50 → implausible

The method must:
- Remove structurally inconsistent values
- Retain rare but valid observations

---
```

$x_{\log} = \log(x)$

```

Apply detection → back-transform

---

### 2. Asymmetric bounds

Lower and upper tails behave differently:

- Learn bounds separately
- Use different thresholds per tail if needed

---

### 3. Bootstrap stability

- Recompute plausible range on resampled data
- Ensure bounds are stable

---
```

Relation to Ensemble Methods

This approach is analogous to:

- Ensemble anomaly detection
- Majority voting classifiers

Key idea:

Combine diverse statistical assumptions to reduce bias and false positives

Future Extensions

- Add multivariate methods (e.g., LOF, Isolation Forest)
- Build concept-level anomaly graphs
- Integrate with OHDSI DataQualityDashboard
- Learn ranges across sites for network-wide validation

Summary

- Use multiple univariate detectors
- Convert outputs to binary flags
- Apply consensus voting
- Derive plausible range from retained values
- Incorporate frequency to improve robustness

This provides a **simple, explainable, and extensible framework** for learning biologically plausible measurement ranges in OMOP CDM.

```

---

# ✅ What to do next

1. Save this as:

plausible_range_consensus.md

2. Add to your repo
3. In Codex, prompt:
> "Read plausible_range_consensus.md and implement this as an R function following HADES sty

---

If you want, I can next:
- Turn this into a fully working HADES-compliant R function
- Or extend it into a QD-compatible check with output tables + thresholds
::contentReference[oaicite:1]{index=1}

```

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It seems that by combining this univariate consensus method, I could learn the min/max for the data and then feed this into the ML LOF method using each value/frequency with attributes from these univariate methods. Is that what you were suggesting?

You're *very close*, but I want to refine the idea so you don't accidentally blunt the power of LOF.

Short answer:

- ✅ Yes—use the consensus method to learn a plausible range and generate features
- ❌ No—don't just feed min/max directly into LOF as primary inputs

The real value is not the bounds themselves—it's the **structure and signals** you derive from them.

💡 What you're proposing (reframed)

You're thinking:

1. Use univariate consensus → learn plausible min/max
2. Feed those into LOF along with value/frequency
3. Detect anomalies

👉 This is directionally correct, but incomplete.

🔥 The key shift (important)

LOF doesn't care about bounds—it cares about **geometry in feature space**

So instead of feeding:

```
(value, frequency, min, max)
```



You want to feed:

how each value relates to the learned structure

✓ Correct architecture (what I recommend)

Step 1 — Learn plausible structure (your consensus method)

Output:

- min_plausible
- max_plausible
- consensus score per value

👉 This step defines:

"what normal looks like"

Step 2 — Transform into features (this is the missing piece)

For each value x , create:

◆ Core features

```
value_scaled
distance_to_min = x - min_plausible
distance_to_max = max_plausible - x
```



◆ Boundary-aware features (very powerful)

```
outside_flag = (x < min_plausible OR x > max_plausible)
distance_to_boundary = min(
    abs(x - min_plausible),
    abs(x - max_plausible)
)
```



◆ Consensus-derived features

```
consensus_score # number of methods flagging it
consensus_flag # majority vote
```



◆ Distribution-aware features

```
z_score
mad_score
percentile
```



◆ Frequency features

```
frequency
log_frequency
support = frequency / total
```



🧩 What this does

Instead of giving LOF raw bounds, you give it:

🔥 A representation of “how weird this value is from multiple perspectives”

🔬 Why this works (important intuition)

LOF detects anomalies when:

A point has **lower density than its neighbors**

So you want feature space where:

- “normal values” cluster tightly
- “implausible values” are isolated

Without transformation (your current idea)

```
(value, frequency, min, max)
```



👉 Problem:

- min/max are constants → no structure
- LOF ignores them

With transformation (recommended)

```
(value_scaled,
 distance_to_boundary,
 consensus_score,
 z_score,
 log_frequency)
```



👉 Now:

- plausible values cluster
- implausible values become isolated

🔥 Even better: treat consensus as a weak label

This is a powerful hybrid approach.

Instead of just using consensus:

Use it as **semi-supervised signal**

Option A: Feature augmentation (simple)

Just include:

consensus_score



Option B: Filtering (stronger)

Train LOF only on:

`x where consensus_score < threshold`

👉 This defines a **clean reference distribution**

Option C: Two-stage detection (best)

1. Consensus → remove obvious outliers
2. LOF → detect subtle anomalies

👉 This mirrors the paper's idea:

- multiple detectors → refined detection

Anomaly_Detection_in_Electronic...

⚠ Important caution

If you over-filter with consensus, LOF loses power

So:

- don't remove too aggressively
- keep borderline points

🧠 Interpretation of your pipeline

What you're building is:

`Univariate methods → consensus → feature engineering → LOF`

This is actually:

🔥 A hybrid statistical + ML anomaly detection system

📊 Example feature vector (final)

For each value:

```
[
  value_scaled,
  z_score,
  mad_score,
  percentile,
  distance_to_min,
  distance_to_max,
  distance_to_boundary,
  consensus_score,
  log_frequency
]
```



🔗 How this connects to your bigger goal

This approach lets you:

- Learn **plausible ranges** (interpretable)
- Detect **subtle anomalies** (LOF)
- Handle **rare but valid values**
- Detect **systematic issues** (via consensus)



Where this gets really interesting (next step)

You can extend this to:

1. Concept-level modeling

- learn ranges per concept

2. Site-level comparison

- detect site-specific deviations

3. Graph layer (like the paper)

- connect concepts with shared anomalies



Bottom line

- ✓ Use consensus to **learn structure and generate features**
- ✗ Don't just pass min/max into LOF
- ✓ Transform everything into **distance + deviation features**
- ✓ Use consensus as:
 - feature
 - filter
 - or weak supervision

If you want, I can next:



Design a **clean R pipeline (HADES style)** that:

- computes consensus
- builds features
- runs LOF
- outputs plausible ranges + anomaly flags

This is very close to something you could plug into DQD or publish.

Sources



Give me the high level blue print so I can review?

Here's the high-level blueprint.

Goal

Learn a **concept-specific biologically plausible range** from observed measurement values, then use that learned structure to support **second-stage anomaly detection with LOF**.

Phase 1 — Prepare the concept-level input

For one measurement concept, build a compact input table with at least:

- measurement value
- frequency of that value

Optional later:

- unit
- care setting
- source/site
- sex/age strata

At the start, keep it concept-specific and unit-harmonized.

Phase 2 — Run a univariate consensus layer

Apply several univariate outlier methods independently to the value distribution for that concept:

- Tukey fences
- quantile thresholds
- z-score
- modified z-score
- generalized ESD

Each method produces its own view of which values are outside the plausible tail behavior.

The important output is **not just a min/max per method**.

The key output is:

- per-value outlier flag from each method
- per-value consensus score
- preliminary consensus-derived plausible range

So this phase gives you a first-pass answer to:

Which values look implausible from multiple statistical perspectives?

Phase 3 — Learn the first-pass plausible range

Use the consensus result to define a retained set of values.

From that retained set, estimate a preliminary plausible range:

- lower plausible bound
- upper plausible bound

These should be derived from the retained values, not by averaging method-specific bounds.

This phase gives you an interpretable artifact:

a learned concept-level plausible min/max

Phase 4 — Convert the consensus output into LOF features

This is the bridge to machine learning.

Do not feed raw min/max into LOF as standalone predictors.

Instead, derive features that describe how each value relates to the learned structure.

Examples of feature families:

Value-position features

- scaled value
- percentile
- z-score
- modified z-score

Boundary features

- distance to lower plausible bound
- distance to upper plausible bound
- distance to nearest boundary
- outside-range flag

Consensus features

- number of univariate methods flagging the value
- majority-vote flag

Frequency features

- raw frequency
- log-frequency
- relative support

This phase transforms the problem from:

“What are the bounds?”

to

“How unusual is this value relative to the learned distribution?”

Phase 5 — Run LOF as a second-stage detector

Use LOF on the engineered feature space.

The purpose of LOF here is not to rediscover the obvious extremes.

The purpose is to identify values that are unusual in a more nuanced way, such as:

- low-support values near the tails
- values borderline by several methods
- values that are not extreme numerically but are isolated in the joint feature space

So LOF becomes a **refinement layer** on top of the univariate consensus layer.

Phase 6 — Combine the outputs

At this stage, you have two signals:

Signal A: consensus-based plausibility

A value is suspicious because multiple univariate methods agree.

Signal B: LOF-based local isolation

A value is suspicious because it is isolated in the engineered feature space.

Combine these into a final review framework such as:

- **high confidence implausible**: consensus outlier and LOF outlier
- **possible implausible**: consensus outlier only
- **possible rare-but-real**: LOF outlier only
- **plausible**: neither flagged

This gives you something much more useful than a single threshold.

Phase 7 — Review and calibrate

Now review the learned range and flagged values against:

- known clinical reference range
- published biologic plausibility
- domain expert opinion
- site/unit expectations

This matters because your target is **biologic plausibility**, not just statistical rarity.

You should expect three nested ideas:

- reference range
- observed common range
- biologically plausible outer range

Your framework is trying to learn the third one.

Phase 8 — Final outputs

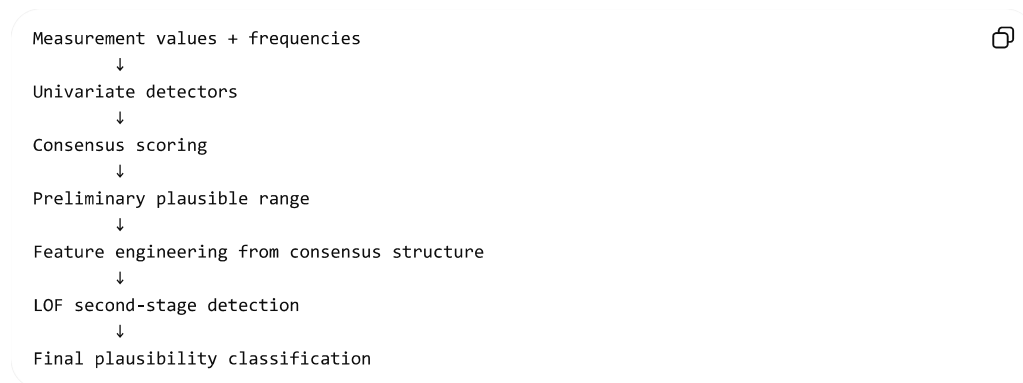
For each concept, produce:

- learned plausible minimum
- learned plausible maximum
- consensus summary by method
- LOF score
- final anomaly classification
- candidate values for manual review

That gives you both:

- an interpretable range artifact
- a ranked anomaly signal

Conceptual architecture



Why this blueprint makes sense

The univariate layer is good at finding obvious tail problems.

The LOF layer is good at finding values that are unusual relative to the overall learned structure.

Together, they give you a hybrid system:

- interpretable enough for data quality work
- flexible enough for subtle anomalies
- extensible to OMOP concepts, sites, and strata

It is very much analogous to the paper's idea of combining multiple detectors and then using a second analytical layer to improve robustness. [Anomaly_Detection_in_Electronic...](#)

Main design decisions for your review

The big things to decide before implementation are:

1. What consensus threshold defines the retained set?
2. Should the final plausible range use min/max or extreme quantiles of the retained set?
3. Which engineered features should go into LOF first?
4. Should frequency be treated as a feature, a weight, or both?
5. How will you validate “biologically plausible” versus merely “statistically common”?

My recommendation

Start simple:

- one concept
- one harmonized unit
- values + frequency only
- consensus layer first
- then LOF with a small feature set:
 - scaled value
 - percentile
 - consensus score
 - distance to nearest plausible boundary
 - log-frequency

That will be enough to tell you whether this architecture is promising before you add age, sex, or care site.

If you want, the next step should be a one-page design spec or HADES-style pseudocode.

Sources



I need to put together a scientific poster to showcase my findings with a specific measurement. The conference organizers suggested the Morrison poster template. Can you advise on how to use that template and how to draw upon the research we've discussed here?

Great—this is a strong story for a poster, and the **Morrison poster template** is actually a perfect fit for what you're doing because it emphasizes *one clear takeaway + minimal text + strong visuals*.

Below is a practical blueprint tailored to your work (consensus + LOF for plausible ranges in OMOP measurement data).

1. How the Morrison poster template works (in plain terms)

The Morrison template (also called “Better Poster”) is built around:

Center (BIG)

One clear takeaway message

Left (Context)

- Background
- Problem
- Methods (light)

➡ Right (Evidence)

- Results
- Figures
- Interpretation

⬇ Bottom (Details)

- Methods details
- References
- Contact

👉 Think:

Headline → Evidence → Supporting details

2. Your core message (this is the most important piece)

Your poster needs a single sentence in the center.

Here are strong options based on your work:

Option A (safe / clear)

A consensus-based ensemble of univariate methods can learn biologically plausible measurement ranges and improve anomaly detection.

Option B (stronger / more novel)

Combining statistical consensus with LOF enables data-driven learning of biologically plausible measurement ranges beyond traditional thresholds.

Option C (OHDSI-focused)

A hybrid consensus + ML approach improves measurement plausibility assessment in OMOP CDM data.

👉 Pick ONE. Make it large. This is what people remember.

3. Left panel (Context + Problem + Approach)

Keep this tight—bullets, not paragraphs.

◆ Background

- Measurement data quality is critical for observational research
- Existing approaches rely on:
 - fixed thresholds
 - reference ranges
- These do not capture **biological plausibility in real-world data**

◆ Problem

- Outliers ≠ implausible values
- Real-world data contains:
 - noise
 - rare but valid observations
 - ETL/unit issues
- Need:

Data-driven method to learn plausible ranges

◆ Approach (high level)

Use a simple diagram here (important).

Data → Univariate Methods → Consensus → Plausible Range → LOF → Refined Anomalies



Label components:

- Tukey, Z-score, MAD, ESD
- Voting / consensus
- Feature engineering
- LOF

👉 This mirrors the **ensemble + second-stage idea from the paper** Anomaly_Detection_in_Electronic...



4. Right panel (Results + Evidence)

This is where your poster wins or loses.

You need **visuals**, not text.

◆ Figure 1 — Distribution + learned range (MUST HAVE)

Plot:

- histogram or density of measurement values
- overlay:
 - clinical reference range
 - learned plausible range (from consensus)

👉 This shows:

"We learned something meaningful"

◆ Figure 2 — Method comparison

Show:

- bounds from each method (boxplot-style or lines)

👉 Demonstrates:

- disagreement across methods
- need for consensus

◆ Figure 3 — Consensus vs LOF (key novelty)

Scatter plot:

- x-axis: value (or percentile)
- y-axis: LOF score

Color:

- consensus outlier
- LOF outlier
- both
- neither

👉 This shows:

- LOF finds subtle anomalies beyond thresholds

◆ Figure 4 (optional but powerful)

Table or small plot showing:

Value	Frequency	Consensus Score	LOF	Classification
👉 Helps reviewers understand behavior				

🔧 5. Bottom section (Methods detail)

Keep it concise but credible.

◆ Data

- OMOP CDM
- single measurement concept
- value + frequency

◆ Univariate methods

- Tukey fences
- Quantiles
- Z-score
- Modified Z-score
- Generalized ESD

◆ Consensus

- voting across methods
- threshold = (e.g., ≥ 3 methods)

◆ Plausible range

- derived from retained values
- extreme quantiles (e.g., 0.1%–99.9%)

◆ LOF

- applied to engineered feature space:
 - value position
 - distance to boundary
 - consensus score
 - frequency

🔧 6. Discussion / Interpretation (short bullets)

- Consensus reduces dependence on any single statistical assumption
- Learned ranges extend beyond clinical reference ranges
- LOF identifies subtle anomalies not captured by thresholds
- Approach is interpretable and extensible to OMOP

⚠️ 7. Limitations (include this—reviewers look for it)

- Single concept evaluation

- Sensitive to unit harmonization
- Assumes unimodal distribution (partially)
- Requires validation against clinical knowledge

8. “Future work” (quick wins)

- Apply across multiple concepts
- Integrate with OHDSI DataQualityDashboard
- Add multivariate/contextual features (age, site, etc.)
- Extend to network-level anomaly detection

9. Design tips for Morrison template

Do:

- Use BIG fonts (seriously—huge)
- Use 3–4 visuals max
- Minimize text
- Use color meaningfully (e.g., red = implausible)

Don’t:

- Write paragraphs
- Over-explain methods
- Show code

10. How this ties back to the paper

If you want to subtly anchor to the literature:

- Your consensus approach ↔ ensemble voting idea
- Your LOF stage ↔ complementary detector
- Your pipeline ↔ multi-stage anomaly detection

👉 One sentence you can include:

“This approach is conceptually aligned with ensemble anomaly detection frameworks that combine multiple detectors to improve robustness.”  Anomaly_Detection_in_Electronic...

Final structure (one-page summary)

CENTER

👉 Big takeaway sentence

LEFT

- Background
- Problem
- Approach diagram

RIGHT

- Distribution + learned range
- Method comparison
- LOF vs consensus plot

BOTTOM

- Methods
- Limitations
- Future work
- References

 If you want next step

I can:

- Draft the **exact poster text** (copy-paste ready)
- Or help you design **specific figures in R (ggplot)** using your data
- Or tailor this to a **specific measurement concept** (e.g., **creatinine**, **potassium**)

This is a strong, novel poster—especially for OHDSI or data quality audiences.

Sources



This is a nice summary thank you. I'd also like to discuss if some of the other ML algorithms I'm considering would be easier to use to start this research vs. LOF? Specifically, information forest and gaussian anomaly detection.

Yes. **Isolation Forest and Gaussian anomaly detection are both easier places to start than LOF** for this kind of early measurement-plausibility research, though they help with different questions.

My practical ranking for a **first prototype** would be:

1. **Gaussian anomaly detection**
2. **Isolation Forest**
3. **LOF**

That is not because LOF is weak. It is because **LOF is the most sensitive to feature design and local density geometry**, which is exactly the part of your project that is still evolving.

Why Gaussian is the easiest starting point

Gaussian anomaly detection is the simplest to explain, implement, and troubleshoot. In the paper you shared, the authors describe it as easy to interpret and computationally light, while also noting that it does not inherently account for temporal order.  Anomaly_Detection_in_Electronic...

For your use case, its strengths are:

- very easy to map back to a learned plausible range
- easy to explain on a poster
- naturally produces lower and upper tails
- useful as a baseline against your univariate consensus range

Why it fits your current stage:

- you are still trying to learn **what plausible range means**
- Gaussian gives you a very transparent first answer
- you can compare it directly to Tukey, quantiles, z-score, modified z-score, and ESD

Its main weakness is also obvious:

- many measurement concepts are skewed, heavy-tailed, or multimodal

- so a plain Gaussian fit may be too crude unless you transform the data, stratify it, or use a more robust variant

For that reason, Gaussian is best framed as:

a simple, interpretable baseline and feature generator

Not necessarily the final model.

Why Isolation Forest is a strong next step

In the paper, Isolation Forest is included as one of the five ensemble detectors because it is effective at isolating unusual observations and can work well in higher-dimensional settings, though the authors note parameter tuning still matters. [Anomaly_Detection_in_Electronic...](#)

For your project, Isolation Forest is attractive because:

- it does **not** require a Gaussian assumption
- it handles nonlinear structure better than simple z-score logic
- it is usually easier to get working than LOF
- it tolerates mixed feature types better than LOF in many practical settings
- it can work well once you start adding:
 - value
 - frequency
 - consensus score
 - distance to learned bounds

Why it may be easier than LOF:

- LOF depends heavily on the notion of a "local neighborhood"
- if your feature space is still immature, LOF can give unstable or hard-to-interpret results
- Isolation Forest is often more forgiving in early exploratory work

For your specific setting, I would think of Isolation Forest as:

a second-stage detector for subtle or structurally unusual values after you establish a plausible-range baseline

Why LOF is harder to start with

LOF can still be very useful, but it is the least forgiving at the beginning.

Its success depends on whether your features create a meaningful local density structure. If you only have crude inputs, LOF often becomes hard to interpret. That is why I was steering you toward using the univariate consensus layer first to create better features before applying LOF.

So LOF is strongest when you already have engineered features like:

- scaled value
- percentile
- distance to lower plausible bound
- distance to upper plausible bound
- consensus score
- log-frequency

Before that, it can be more work than it is worth.

A good research progression

I would not frame this as choosing one winner immediately. I would structure the research in stages.

Stage 1: transparent statistical baseline

Use your univariate consensus method to learn a plausible range.

Then compare that to:

- Gaussian anomaly detection on raw or transformed values
- maybe Gaussian on frequency-aware representations

This gives you an interpretable starting point.

Stage 2: easy ML extension

Add Isolation Forest using a small feature set such as:

- scaled value
- percentile
- log-frequency
- consensus score
- distance to nearest plausible boundary

This tests whether ML adds signal beyond threshold-based methods.

Stage 3: local-density refinement

Only then bring in LOF with the same feature set and compare:

- what LOF finds that Isolation Forest does not
- whether those points are clinically or biologically interesting

That progression gives you a clean scientific narrative:

start simple, establish plausibility, then test whether more flexible ML methods add value

If your goal is a poster or early manuscript

From a communication standpoint, this is the easiest story:

- **Consensus univariate methods** learn an initial plausible range
- **Gaussian** provides an interpretable baseline
- **Isolation Forest** tests whether flexible ML improves anomaly detection
- **LOF** is an optional refinement once feature engineering matures

That is much easier to explain than starting with LOF alone.

My recommendation

If you want the easiest entry point, start with:

- **Consensus range**
- **Gaussian anomaly detection**
- **Isolation Forest**

Then add LOF only after you have a stable feature space.

So the answer is:

- **Gaussian** is easiest to start
- **Isolation Forest** is probably the easiest ML method to add next
- **LOF** is likely the hardest of the three to use well at the beginning

A very reasonable first comparison study would be:

1. univariate consensus only
2. consensus + Gaussian
3. consensus + Isolation Forest
4. consensus + LOF

That would let you ask a very clean question:

Does increasingly flexible anomaly detection improve learning of biologically plausible measurement bounds?

If you want, I can sketch a side-by-side comparison table for **Gaussian vs Isolation Forest vs LOF** specifically for OMOP measurement plausibility research.

Sources



Please sketch that table

Here's a **comparison table tailored to your exact use case**: learning biologically plausible ranges for OMOP measurement data using consensus + ML.

 Comparison: Gaussian vs Isolation Forest vs LOF

(for measurement plausibility + consensus integration)

Dimension	Gaussian Anomaly Detection	Isolation Forest	Local Outlier Factor (LOF)
Conceptual model	Assumes data follows a probability distribution (typically normal)	Isolates anomalies via random partitioning	Detects points with lower local density than neighbors
Ease of use (early research)	★★★★★ Very easy	★★★★ Easy	★★★ Harder
Interpretability	★★★★★ High (mean, SD, tails)	★★★ Moderate	★★ Low-moderate
Alignment with plausible range estimation	★★★★★ Strong (direct min/max from tails)	★★★★ Moderate (implicit)	★★★ Weak (no explicit bounds)
Handles skewed distributions	★★★ Weak (unless transformed)	★★★★ Good	★★★★ Good
Handles multimodal data	★★ Weak	★★★★ Good	★★★★★ Strong
Sensitivity to feature engineering	★★ Low	★★★ Moderate	★★★★★ High
Uses frequency naturally	★★★ Limited (needs weighting)	★★★★ Good	★★★★ Good
Detects extreme outliers	★★★★★ Strong	★★★★★ Strong	★★★★ Moderate
Detects subtle/local anomalies	★★★ Weak	★★★★ Moderate	★★★★★ Strong
Robust to noisy real-world data	★★★ Moderate	★★★★★ Strong	★★★★ Moderate
Hyperparameter tuning burden	★★ Low	★★★ Moderate	★★★★ Moderate-high
Computational cost	★★ Low	★★★ Moderate	★★★★ Higher (for large N)

 How each fits into *your* pipeline

Role in your workflow	Gaussian	Isolation Forest	LOF
Learn initial plausible range	✅ Best fit	⚠️ Indirect	❌ Not suited
Validate consensus range	✅ Strong	✅ Good	⚠️ Limited
Detect ETL/unit errors	✅ Strong (extremes)	✅ Strong	⚠️ Mixed

Role in your workflow	Gaussian	Isolation Forest	LOF
Detect rare but plausible values	⚠️ Weak	✅ Good	✅ Strong
Detect subtle anomalies near boundaries	❌ Weak	⚠️ Moderate	✅ Best
Second-stage refinement after consensus	⚠️ Limited	✅ Strong	✅ Strong (if features good)

 Practical recommendation (based on this table)

Start here (fast + interpretable)

- Consensus + Gaussian
 - Learn plausible range
 - Validate against known biology

Add next (strong ML baseline)

- Consensus + Isolation Forest
 - Capture nonlinear / irregular patterns
 - Handle frequency + engineered features well

Add last (advanced refinement)

- Consensus + LOF
 - Only after feature space is well-designed
 - Use to detect subtle density-based anomalies

 What each method will *actually* show you

Gaussian

- “What values are statistically extreme?”
- Produces:
 - clear lower/upper tails
 - easy-to-explain thresholds

👉 Best for your **poster + initial narrative**

Isolation Forest

- “What values are structurally unusual?”
- Produces:
 - anomaly scores
 - robust detection across mixed features

👉 Best for showing **ML adds value beyond thresholds**

LOF

- “What values don’t belong to any local neighborhood?”
- Produces:
 - fine-grained anomaly detection

- subtle boundary violations

👉 Best for **refinement and deeper research (not first step)**

⚠️ **Key insight (important)**

These methods are **not competitors**—they answer different questions.

Question	Best method
What is the plausible range?	Gaussian + consensus
Are there unusual values globally?	Isolation Forest
Are there locally inconsistent values?	LOF

🔗 **Suggested evaluation framing (for your poster/paper)**

You can structure your results like this:

1. **Consensus range vs Gaussian**
 - Do they agree?
 - Where do they differ?
2. **Add Isolation Forest**
 - What new anomalies appear?
 - Are they clinically meaningful?
3. **Add LOF**
 - What subtle anomalies are detected?
 - Do they reflect real data issues?

✅ **Bottom line**

- **Gaussian** → best starting point (simple, interpretable, range-focused)
- **Isolation Forest** → best early ML extension (robust, flexible)
- **LOF** → best advanced method (but requires careful feature design)

If you want next step, I can:

- Map this table directly into a **poster figure (very compelling visual)**
- Or design a **side-by-side experiment plan with metrics (precision/recall vs expert review)**

📄 👍 🗨️ ⬆️ ↺ ⋮

Map this into a poster figure but let's leave LOF out of this scenario - I think it will be too hard to fit into the poster.

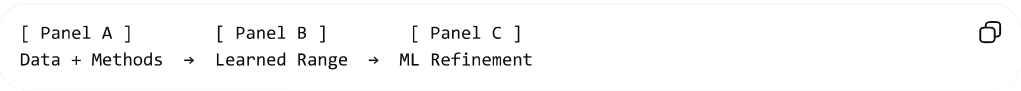
Great call—removing LOF will make your poster **cleaner, more interpretable, and easier to defend**. You can still tell a strong story with:

Consensus (statistical ensemble) + Gaussian + Isolation Forest

Below is a **poster-ready figure design** you can implement directly (e.g., in ggplot2 or PowerPoint).

 Poster Figure: “Learning Plausible Ranges with Consensus + ML”

 Overall layout (single figure with 3 panels)



 Title for the figure:

“Consensus-based range learning with ML refinement improves detection of implausible measurement values”

 Panel A — Methods overview (left)

Goal

Show the pipeline clearly and simply

Visual (diagram)



- Add small annotations:
- “Multiple statistical assumptions”
 - “Voting reduces method bias”
-  Keep this minimal—icons or simple boxes

 Panel B — Learned plausible range (center, main visual)

This is your most important panel

Plot: Distribution of values

- Use:
- histogram or density curve
- Overlay:
1. Clinical reference range (dashed line)
 2. Consensus plausible range (solid line)
 3. Optional: Gaussian range (different color)

Example visual encoding

Element	Style
Histogram	light gray
Clinical range	dashed black
Consensus range	thick blue

Element	Style
Gaussian range	thin green

Add labels directly on plot:

- “Reference range”
- “Consensus-derived plausible range”
- “Gaussian estimate”

Key message (small caption)

Consensus methods learn a plausible range that extends beyond traditional clinical thresholds while remaining biologically consistent.

 Panel C — ML refinement (right)

Goal

Show how Isolation Forest adds value

Plot: Scatter or strip plot

Option A (best):

- x-axis: measurement value
- y-axis: Isolation Forest score

Color points by:

- consensus inlier
- consensus outlier
- IF outlier

Color scheme (important)

Category	Color
Plausible (consensus inlier)	gray
Consensus outlier	red
IF-only outlier	orange
Both (strong anomaly)	dark red

What this shows

- IF identifies:
 - subtle anomalies near boundaries
 - low-frequency unusual values

Caption

Isolation Forest identifies structurally unusual values beyond consensus-based thresholds.

 Optional small inset (very effective)

Tiny table:

Value	Freq	Consensus	IF	Interpretation
6.5	low	inlier	outlier	rare but plausible
50	low	outlier	outlier	implausible
2.1	high	inlier	inlier	normal

 This helps reviewers immediately “get it”

 Design tips (critical for Morrison template)

Make Panel B dominant

- ~50% of visual weight
- largest font
- central position

Keep Panel A minimal

- diagram only
- no paragraphs

Use color meaningfully

- blue = learned structure
- red = implausible
- gray = normal

 Narrative flow (how people read it)

1. **Center panel** → “Oh, they learned a range”
2. **Left panel** → “Ah, they used multiple methods”
3. **Right panel** → “ML finds additional anomalies”

 This aligns perfectly with Morrison design principles

 Subtle tie to the paper

You can include a small note:

“Inspired by ensemble anomaly detection frameworks combining multiple detectors for robust signal identification”  Anomaly_Detection_in_Electronic...

 Final takeaway (what your figure communicates)

Your figure should make this obvious without reading text:

“We combine multiple statistical methods to learn a plausible range, and then use machine learning to detect additional anomalies beyond that range.”

👍 If you want next step

I can:

- Generate **ggplot2** code for all 3 panels
- Or mock up a **PowerPoint/Keynote** layout
- Or tailor this to a **specific measurement** (e.g., **potassium, creatinine**)

This figure will likely be the centerpiece of your poster—worth getting just right.

Sources



I'd like for you to write up the methods described here so that I can share them with my research team. I'd also like for you to write the anticipated results. Since I do not have the results just yet, what I'd like for the results to include are what we think we'd find by applying the methods.

Below is a **draft Methods + Anticipated Results** section you can share directly with your team (and later adapt for a poster or manuscript). It's written in a scientific tone but kept concise and structured.

Methods

Study Objective

The objective of this study is to develop and evaluate a data-driven framework for learning **biologically plausible value ranges** for measurement concepts in the OMOP Common Data Model (CDM). The approach combines multiple univariate statistical outlier detection methods with machine learning to improve identification of implausible measurement values in real-world observational data.

Data Source and Cohort Definition

We selected a single measurement concept from an OMOP CDM-formatted database. For this concept, we extracted:

- Measurement value (`value_as_number`)
- Frequency of occurrence of each value

To simplify the initial evaluation, analyses were conducted:

- At the concept level (no stratification by age, sex, or care setting)
- Assuming harmonized units
- Using aggregated value-frequency pairs rather than patient-level longitudinal data

Univariate Outlier Detection Methods

We applied five commonly used univariate outlier detection methods to the distribution of measurement values:

1. **Tukey fences** (interquartile range-based thresholds)
2. **Quantile thresholds** (extreme percentile cutoffs)
3. **Z-score thresholds** (mean and standard deviation-based)
4. **Modified Z-score** (median absolute deviation-based)
5. **Generalized Extreme Studentized Deviate (ESD)** test

Each method independently classified values as inliers or outliers based on its respective statistical assumptions.

Consensus-Based Outlier Detection

To reduce dependence on any single statistical assumption, we implemented a **consensus (voting) approach** across the five methods.

For each measurement value:

- A binary outlier flag was assigned by each method
- A **consensus score** was calculated as the number of methods flagging the value as an outlier

Values were classified based on a predefined threshold (e.g., ≥ 3 methods):

- **Consensus outliers**: flagged by multiple methods
 - **Consensus inliers**: not consistently flagged
-

Learning Plausible Value Ranges

A preliminary **plausible value range** was derived from the subset of values classified as consensus inliers.

To ensure robustness to residual noise and extreme values, the plausible range was defined using extreme quantiles of the retained distribution (e.g., 0.1th and 99.9th percentiles), rather than raw minimum and maximum values.

This approach provides a data-driven estimate of biologically plausible bounds that is less sensitive to outliers and distributional assumptions.

Gaussian Anomaly Detection

As a baseline statistical model, we applied **Gaussian anomaly detection** to the measurement values.

- The distribution of values was modeled using mean and standard deviation
- Values with low probability under the fitted distribution were flagged as anomalous

This method provides a simple, interpretable comparison to the consensus-derived plausible range and helps assess how well standard parametric assumptions capture the observed data.

Isolation Forest

To identify more complex and non-linear anomalies, we applied **Isolation Forest** as a machine learning-based anomaly detection method.

Isolation Forest operates by recursively partitioning the data and assigning anomaly scores based on how easily observations are isolated.

Feature Engineering

Each measurement value was represented using a set of engineered features derived from the consensus and distributional structure:

- Scaled measurement value
- Percentile within the distribution
- Distance to lower and upper plausible bounds
- Distance to nearest plausible boundary
- Consensus score (number of methods flagging the value)
- Log-transformed frequency

These features capture both the position of each value within the distribution and its structural relationship to the learned plausible range.

Integration of Statistical and ML Signals

Final anomaly characterization was based on combining:

- **Consensus-based classification** (statistical agreement across methods)
- **Isolation Forest anomaly scores** (structural deviation in feature space)

Values were categorized into:

- **High-confidence implausible:** flagged by both consensus and Isolation Forest
- **Potentially implausible:** flagged by consensus only
- **Structurally unusual:** flagged by Isolation Forest only
- **Plausible:** not flagged by either approach



Anticipated Results

Consensus-Based Plausible Range

We expect the consensus-based approach to produce a **plausible value range that differs from both clinical reference ranges and individual statistical thresholds**.

Specifically:

- The learned range will likely extend beyond standard clinical reference ranges, reflecting real-world variability
- Compared to individual methods, the consensus approach will produce **more stable and robust bounds**, reducing sensitivity to distributional assumptions
- Extreme implausible values (e.g., unit errors or data entry artifacts) will be consistently excluded across methods

Comparison to Gaussian Anomaly Detection

We anticipate that Gaussian-based thresholds will:

- Provide a reasonable approximation of the central distribution
- Be sensitive to skewness and heavy tails in measurement data
- Either over-restrict or over-extend plausible ranges depending on distribution shape

As a result:

- Gaussian-derived bounds may diverge from consensus-based bounds, particularly in skewed or multimodal distributions
- This comparison will highlight the limitations of relying on parametric assumptions for real-world observational data

Isolation Forest Findings

We expect Isolation Forest to identify **additional anomalies not captured by univariate methods**, including:

- Values near the boundaries of the plausible range that are rare or structurally isolated
- Low-frequency values that are not extreme in magnitude but are inconsistent with the overall distribution
- Subtle irregularities that reflect data quality issues not detectable through threshold-based methods

These findings will demonstrate that machine learning methods can provide **complementary signals beyond traditional statistical approaches**.

Combined Interpretation

By integrating consensus-based and machine learning-based approaches, we anticipate identifying distinct categories of measurement values:

- **Clearly implausible values:** extreme outliers consistently flagged across methods
- **Borderline values:** values near distribution boundaries with partial agreement
- **Rare but plausible values:** low-frequency values identified by machine learning but not statistical consensus
- **Common plausible values:** high-frequency values within the learned range

Implications for Data Quality Assessment

Overall, we expect this framework to:

- Improve identification of implausible measurement values in OMOP CDM data
- Provide a **data-driven alternative to fixed thresholds and reference ranges**
- Offer a scalable and interpretable approach for integration into data quality tools such as the OHDSI DataQualityDashboard
- Enable more nuanced differentiation between **biologically implausible values and rare but valid observations**

✓ Summary

This study proposes a hybrid framework that:

- Uses **consensus across multiple statistical methods** to learn plausible ranges
- Applies **machine learning (Isolation Forest)** to detect additional anomalies
- Combines both approaches to produce a more robust and interpretable assessment of measurement plausibility

If you'd like, I can next:

- tighten this into a **poster-ready version (very concise)**
- or convert it into a **manuscript-style Methods section with citations**
- or align it explicitly with **OHDSI DQD terminology and outputs**



Can you write this in paragraph style vs. the bullet points? I'd like to be able to place it into an abstract.

Here is a **paragraph-style version suitable for an abstract**, combining methods and anticipated results into a cohesive narrative:

We developed a data-driven framework to learn biologically plausible value ranges for measurement concepts within the OMOP Common Data Model (CDM) by integrating multiple univariate statistical methods with machine learning–based anomaly detection. Using a single measurement concept, we analyzed aggregated measurement values and their frequencies, assuming harmonized units and no initial stratification by demographic or care setting variables. Five univariate outlier detection methods—Tukey fences, quantile thresholds, Z-scores, modified Z-scores, and the generalized extreme Studentized deviate (ESD) test—were applied independently to characterize the distribution of measurement values. To reduce dependence on any single statistical assumption, we implemented a consensus-based voting approach in which each value received an outlier score corresponding to the number of methods that flagged it as anomalous. Values not consistently identified as outliers were retained to estimate a preliminary plausible range, defined using extreme quantiles of the retained distribution to ensure robustness to noise and distributional irregularities. As a parametric baseline, Gaussian anomaly detection was applied to model the overall distribution and identify values with low probability under a normality assumption. To capture more

complex and non-linear patterns, we then applied Isolation Forest using engineered features that describe each value's position relative to the learned distribution, including scaled value, percentile, distance to the plausible range boundaries, consensus score, and log-transformed frequency. Final anomaly characterization combined signals from both statistical consensus and machine learning, enabling classification of values into plausible, potentially implausible, and structurally unusual categories.

We anticipate that the consensus-based approach will yield a plausible value range that is more stable and robust than thresholds derived from any single method and that extends beyond traditional clinical reference ranges while remaining biologically consistent. Compared to Gaussian-based thresholds, which are sensitive to distributional assumptions, the consensus-derived range is expected to better accommodate skewed and heavy-tailed measurement distributions commonly observed in real-world data. Isolation Forest is expected to identify additional anomalies not captured by univariate methods, particularly low-frequency or structurally isolated values near distribution boundaries, thereby providing complementary information beyond threshold-based approaches. Together, these methods are expected to improve differentiation between implausible measurement values and rare but valid observations, supporting a more nuanced and scalable approach to data quality assessment in observational health data and offering potential integration with tools such as the OHDSI DataQualityDashboard.

If you want, I can also:

- shorten this to a **strict 250–300 word conference abstract**, or
- tailor it to a **specific measurement (e.g., potassium, creatinine)** for stronger impact.

📄 👍 💬 ↗️ ↺ ...

I'm redoing my project with the goal being finding the right minimum/maximum values for a specific measurement value (and unit) using univariate methods, information forest and GAD. I'm unsure how to compare the outputs of these methods. Assume for IF and GAD, the features in my model will be the gender and age (5-year age midpoints). Any advice on how to approach this?

This is a really good refinement of your problem—and you're now asking the *right* question:

How do I compare fundamentally different types of outputs to learn a single “best” plausible min/max?

Because now you have:

- **Univariate methods** → explicit bounds
- **Gaussian (GAD)** → probabilistic density
- **Isolation Forest (IF)** → anomaly scores (no bounds)

These are *not directly comparable outputs*, so the trick is:

🔥 **You need to project all methods into a common evaluation space**



1. Reframe the problem (this unlocks everything)

Instead of asking:

“Which method gives the best min/max?”

Ask:

“Which values are plausibly inliers vs implausible outliers?”

Then derive min/max *from that agreement*.

 2. Normalize all methods to a common representation

You want every method to produce:

Plausible vs Implausible classification per value



 Univariate methods (already there)

You already have:

- bounds → classify values
- output: binary (inlier / outlier)

 Gaussian (GAD)

Convert probability → classification:

Options:

- z-score threshold (e.g., $|z| > 3$)
- tail probability threshold (e.g., $p < 0.001$)

 Output:

inlier / outlier



 Isolation Forest

Convert anomaly score → classification:

Options:

- contamination parameter (e.g., 0.1%)
- score percentile cutoff

 Output:

inlier / outlier



 3. Now everything is comparable

You now have a matrix like:

Value	Tukey	Quantile	Z	MAD	ESD	GAD	IF
x_1	0	0	0	0	0	0	0
x_2	1	1	1	0	1	1	0
x_3	1	1	1	1	1	1	1

(1 = outlier)

 4. Build a unified consensus score

```
consensus_score = sum(outlier flags)
```



Now you can:

- define thresholds:
 - $\geq 5 \rightarrow$ strong implausible
 - 3–4 $\rightarrow$ borderline
 - $\leq 2 \rightarrow$ plausible

5. Derive min/max from agreement (key step)

Instead of picking a method:

 Use the **retained plausible values**

```
plausible_values = values where consensus_score < threshold
```



Then:

```
min_plausible = lower quantile(plausible_values)
max_plausible = upper quantile(plausible_values)
```



6. Where age + gender come in (your key addition)

This is where your design becomes more powerful than before.

Option A (recommended first): stratified modeling

Run the entire pipeline separately for:

- gender
- age groups (5-year bins)

 Output:

```
min/max per (age group × gender)
```



Option B: include as features in IF/GAD

You proposed:

- age midpoint
- gender

This works well for IF, but less naturally for GAD unless you model:

- conditional distributions
- or separate Gaussians per stratum

Key insight

Age and gender are **not just features**—they define different distributions

So:

 Best practice:

- use them for **stratification first**
- then optionally include them as features

 7. How to compare methods (this is your core question)

Now that everything is aligned, you can compare along 3 axes:

1. Agreement with consensus

For each method:

- how often does it agree with consensus?
- does it over-flag or under-flag?

2. Impact on learned range

Compute:

```
range_method = [min_m, max_m]
range_consensus = [min_c, max_c]
```

Compare:

- width
- overlap
- asymmetry

3. Stability

Bootstrap your data:

- recompute min/max
- see which method is stable

 8. The most important comparison (don't skip this)

Instead of comparing methods directly:

 Compare their effect on the final plausible range

Example:

Method	Min	Max	Width
Tukey	2.1	8.5	narrow
Z-score	1.8	9.2	wider
GAD	1.5	10.5	very wide
IF-derived	implicit	implicit	N/A
Consensus	1.9	9.0	balanced

 This is what reviewers care about.

 9. Clean workflow (what I recommend)

Step 1

Run all methods → convert to binary flags

Step 2

Build consensus score

Step 3

Define plausible values

Step 4

Compute min/max

Step 5

Compare:

- individual methods vs consensus
- stratified vs non-stratified

10. Common pitfalls (important)

Comparing raw outputs

- IF score vs z-score → meaningless comparison

Using IF to define bounds directly

- IF doesn't produce interpretable min/max

Ignoring stratification

- age/gender effects can dominate signal

11. Recommended study framing

This would be a strong narrative:

"We evaluated whether combining univariate statistical methods with Gaussian and Isolation Forest anomaly detection improves estimation of biologically plausible measurement ranges."

Then show:

1. Individual methods disagree
2. Consensus stabilizes range
3. IF adds detection of subtle anomalies
4. Stratification improves biological realism

Bottom line

- Convert all methods → **binary outlier flags**
- Use **consensus as the unifying layer**
- Learn min/max from **retained values**
- Treat IF/GAD as **complementary signals**, not direct competitors
- Use **age/gender for stratification first**, features second

If you want, I can next:

 Turn this into a **formal evaluation framework (metrics + plots)**

 Or write **R pseudocode** aligned with **HADES style** for this exact pipeline

     ...

The methods in my anomaly detection package will take, as inputs, the `data.frame` with the measurement value (think systolic blood pressure reading), the frequency of this value, the age-midpoint and value. It will then return the same `data.frame` with a binary to indicate if the method indicated that it is an outlier. Given this information, would you adjust anything from your previous response?

Yes — I'd adjust it in a few important ways.

Given your package interface, all methods already return a **row-level binary outlier flag on the same input table**, which simplifies the comparison problem a lot. You no longer need to force very different outputs into a common space later — your package is already doing that.

What changes

The main shift is this:

Your primary comparison unit is no longer the raw method output. It is the per-row outlier decision.

So instead of comparing:

- bounds from univariate methods
- scores from IF
- probabilities from GAD

you can compare:

- **which rows each method flags**
- **how those row-level flags translate into a learned min/max**

That makes the overall framework cleaner.

Updated framing

Your input table sounds like it has, per row:

- measurement value
- frequency of that value
- age midpoint
- sex/gender

and each method returns the same table plus:

- `isOutlier_<method>`

That means your workflow should become:

1. Run each method on the same input table
2. Append one binary outlier column per method
3. Compare methods at the row level
4. Build a consensus flag
5. Learn the final min/max from rows not flagged as outliers

The most important adjustment

Previously I suggested that age and gender might be better used for stratification first. With your package design, I would revise that to:

Use **both approaches conceptually**, but start with **age midpoint and gender as explicit model inputs**, since that is how your methods are already defined.

That means your immediate question is not “should I stratify first?” but rather:

“Do methods behave sensibly when age midpoint and gender are included in the same row-level representation?”

For a first version, that is completely reasonable.

How I would now compare the methods

Since every method returns a binary flag on the same rows, I would compare them in three layers.

1. Agreement at the row level

Build a matrix like this:

value	frequency	ageMidpoint	gender	tukey	quantile	zScore
120	54	47.5	M	0	0	0
260	1	72.5	F	1	1	1

Then compute:

- consensus score = number of methods flagging the row
- consensus flag = binary based on threshold, for example ≥ 4

This becomes your common comparison layer.

2. Effect on retained rows

For each method, look at the rows it leaves in.

That is more meaningful than comparing internal mechanics, because your end goal is:

Which rows remain plausible enough to define the min/max?

3. Effect on learned min/max

For each method individually, and then for consensus, compute:

- lower plausible bound
- upper plausible bound

from the rows that are not flagged.

This is probably the most important comparison for your project.

What I would keep from the previous response

I would still keep this core principle:

Do not try to compare IF and GAD to univariate methods based on their native score scales.

Because your package already converts everything to a binary outlier flag, the right comparison is downstream:

- row-level agreement
- disagreement patterns
- final learned bounds

That remains true.

What I would newly emphasize

Because you are passing value , frequency , age midpoint , and gender into all methods, I would now be very explicit about this design question:

Are the univariate methods truly using those covariates, or are they only using value/frequency while the multivariate methods use all fields?

This matters a lot.

If Tukey, quantile, z-score, modified z-score, and ESD are still fundamentally acting on the marginal distribution of value, while IF and GAD use age/gender context, then you are comparing:

- **marginal outlier detection**
versus
- **context-aware outlier detection**

That is fine, but you should describe it honestly that way.

In fact, that can become part of your scientific question:

Do contextual methods yield different plausible bounds than value-only methods?

One recommendation I would add

Because each row represents a value-frequency-age-gender combination, I would separate two concepts:

Global plausibility

Is this systolic blood pressure value implausible for the concept overall?

Contextual plausibility

Is this value implausible given age midpoint and gender?

That suggests two possible outputs:

- `isOutlier_global`
- `isOutlier_contextual`

You do not need to implement both immediately, but conceptually it is helpful. Otherwise you may mix:

- “260 is globally extreme”
with
- “145 is unusual only for a young subgroup”

Those are different scientific claims.

How to derive min/max from this structure

Given your interface, I would recommend:

For each method:

1. Filter to rows with `isOutlier_method == 0`
2. Expand by frequency conceptually, or use weighted quantiles
3. Derive:
 - lower plausible bound
 - upper plausible bound

Then do the same for consensus.

This gives you a clean summary table like:

method	lowerBound	upperBound	excludedRows	excludedWeight
tukey	85	210	12	18
gad	82	220	15	22

method	lowerBound	upperBound	excludedRows	excludedWeight
if	88	205	20	25
consensus	86	208	14	20

That is probably the best way to compare outputs for your team and for a poster.

One caution about GAD and IF with your inputs

Because you are using:

- value
- frequency
- age midpoint
- gender

I would expect:

GAD

to be more sensitive to how you encode gender and how close the joint distribution is to Gaussian.

IF

to be more forgiving and likely easier to use as a contextual method.

So I would not expect GAD and IF to define bounds directly in the same way as univariate methods. I would treat them as:

row-level contextual filters whose impact is assessed through the retained-value range.

That is the clean comparison.

Revised recommendation

Given your package design, my updated recommendation is:

- Keep all methods returning binary row-level flags
- Compare methods using:
 - row-level agreement
 - consensus score
 - retained-value min/max
- Treat univariate methods as mostly **value-based**
- Treat IF/GAD as potentially **context-aware**
- Derive the final plausible range from the rows retained by consensus, not from any one method alone

Bottom line

Yes, I would adjust the framing, but not the core strategy.

The main adjustment is:

Since all methods return the same row-level binary output, your comparison framework should center on **agreement across flagged rows and the resulting plausible bounds**, rather than on the native mechanics of each method.

That makes your project much cleaner.

If you'd like, I can turn this into a compact design note for your team with a proposed comparison table and evaluation workflow.



At the moment, I'm only passing in the value & frequency to the univariate methods and reading your response, this feels like a mistake. Should I run the univariate methods on each strata of age/gender?

Yes — **stratifying the univariate methods by age/gender is a sensible next step**, but I would do it **selectively**, not automatically for every concept.

The reason is simple:

a value can be statistically unusual overall, but perfectly plausible within a demographic stratum.

For something like **systolic blood pressure**, age matters a lot, and sex may matter somewhat. So if you only run univariate methods on the pooled value distribution, you risk learning bounds that are really an average of several different subdistributions rather than a true plausible range.

What this means in practice

Right now your univariate methods are effectively answering:

"Is this value unusual for the concept overall?"

If you stratify by age/gender, they instead answer:

"Is this value unusual for this age/gender subgroup?"

That is often closer to biologic plausibility.

My recommendation

I would not replace the pooled analysis entirely. I would run **both**:

- a **global** univariate analysis on all values for the concept
- a **stratified** univariate analysis within age/gender groups

That gives you two complementary views:

- **global plausibility**: catches obviously extreme values
- **contextual plausibility**: catches values unusual for a subgroup

This is better than jumping straight to stratified-only, because small strata can get unstable.

Why not always stratify?

Because there are tradeoffs.

Benefits

Stratification can:

- reduce mixing of distinct distributions
- produce more realistic lower/upper bounds
- better reflect physiology for age-sensitive measurements

Risks

Stratification can also:

- create very small cells
- make Tukey, z-score, ESD, and quantile thresholds unstable
- overfit noise in sparse groups

So the real answer is:

Yes, stratify when the measurement distribution is meaningfully age- or sex-dependent and the strata have enough data.

A practical way to do it

For a first study, I'd suggest:

- **gender:** stratify directly
- **age:** use broader bands first, not too many 5-year bins unless you have a lot of data

For example:

- 0–17
- 18–39
- 40–64
- 65+

Or, if your dataset is large enough:

- 18–29
- 30–44
- 45–59
- 60–74
- 75+

Then, only move to 5-year bins if sample sizes remain strong.

What I would avoid

I would avoid running univariate methods on tiny age × gender cells where:

- the number of distinct values is low
- the frequency distribution is sparse
- ESD or z-score assumptions become brittle

That can make the learned min/max look precise but actually be noisy.

A good compromise

A strong design would be:

1. Run univariate methods on the full concept distribution.
2. Run them again within age/gender strata.
3. Compare the resulting bounds.

Then ask:

- Do stratified bounds differ materially from pooled bounds?
- Are those differences clinically sensible?
- Do the stratified results stabilize or fragment the signal?

That gives you an empirical basis for deciding whether stratification helps.

How this fits with IF and GAD

This actually makes your framework cleaner:

- **univariate methods:** run globally and by stratum
- **IF/GAD:** use age and gender directly as model inputs

Then you can compare:

- pooled statistical bounds
- stratified statistical bounds
- context-aware ML flags

That is a much stronger story than pooled-only univariate methods.

My bottom-line advice

For measurements like systolic blood pressure, **yes, I would add age/gender stratification to the univariate methods**, but I would do it carefully:

- keep the pooled version
- start with coarse strata
- only use finer bins if data support it
- compare stratified versus pooled bounds as part of the analysis

So it is not a mistake to have started with value + frequency only. That was a reasonable first pass. But for the next iteration, stratified univariate analysis is probably the right move.

If you want, I can sketch a simple decision rule for when a concept should be analyzed pooled versus stratified.



Can you suggest age bands that are defensible for systolic blood pressure?

Thought for 9s >

Yes. For **systolic blood pressure**, I'd use **clinically motivated age bands rather than arbitrary 5-year bins**, especially for a first pass.

A defensible starting scheme is:

- 3–12 years
- 13–17 years
- 18–39 years
- 40–64 years
- 65–79 years
- 80+ years

Why this works:

For **children under 13**, blood pressure is interpreted using **age-, sex-, and height-based percentiles**, so finer pediatric stratification is usually needed and a single pooled pediatric band is still a simplification. For **adolescents 13+**, the AAP guideline aligns the simplified threshold table with adult-style cut points, making 13–17 a natural bridge band. NHLBI, NIH +1

For **adults**, 18–39, 40–64, 65–79, and 80+ are defensible because they reflect meaningful changes in blood-pressure epidemiology and guideline framing. Major adult guidance uses age-related considerations for older adults, and European guidance explicitly distinguishes **65–79** from **80+** for systolic targets, which gives those two older-adult bands a strong clinical rationale. American Colleg... +1

If your dataset is **adult-only**, I would simplify to:

- 18–39
- 40–64
- 65–79
- 80+

That is probably the cleanest and easiest-to-defend adult scheme. American Colleg... +1

If you need **more sample size per cell**, a coarser adult scheme is also reasonable:

- 18–44
- 45–64
- 65–79
- 80+

I would avoid starting with 5-year bins unless you have a very large sample, because your univariate methods can become unstable in sparse strata.

Two cautions matter here.

First, if you include **children**, systolic BP is not really well handled by age bands alone because pediatric interpretation also depends on **height and sex**. So for a scientifically strong analysis, I would either analyze

adults only first or treat pediatrics separately. NHLBI, NIH +2

Second, these bands are good for **modeling plausibility**, but they are not the same thing as the standard adult BP category cutoffs like normal, elevated, stage 1, and stage 2 hypertension. Those are value categories, not age strata. www.heart.org +1

My practical recommendation would be:

- If your study is **adult-focused**, use **18–39 / 40–64 / 65–79 / 80+**
- If your study includes pediatrics, separate them first into **3–12 / 13–17 / adults**, and ideally handle the pediatric group with extra caution because of the height dependency. NHLBI, NIH +1

If you want, I can turn this into a short “Methods justification” paragraph you can paste into your abstract or poster.

       Sources

So let's go back to the other response where you laid out a plan. If I've got it right, I should run the univariate methods on the global set of measurement values and report out the min/max for each method. I can then do the same univariate outlier detection methods on data stratified by age buckets and gender if the data are large. Then I can run IF and GAD using the age midpoint and gender features. If this is right, how do I then compare across what I've done to come to some conclusions?

Yes — that is the right overall plan.

The main remaining question is not “how do I run the methods?” but:

What is the comparison framework that lets me draw conclusions across global, stratified, IF, and GAD results?

The cleanest way to do this is to treat everything as contributing to **three levels of evidence**:

1. **Method-specific bounds**
2. **Agreement/disagreement about flagged values**
3. **A final recommended plausible range**

That gives you a way to compare unlike methods without forcing them to be identical.

1. Keep the outputs separated into two families

You will have:

Global univariate results

For each method, on the full concept distribution:

- lower bound
- upper bound
- number of values flagged
- total frequency excluded

Stratified univariate results

For each method, within each age band × gender stratum:

- lower bound
- upper bound
- number of values flagged
- total frequency excluded

IF and GAD results

These are not naturally “bound methods” in the same way, even if you can derive flagged rows from them. So treat them as:

- context-aware outlier detectors
- methods that tell you which rows look implausible given value, frequency, age midpoint, and gender

That distinction is important. The univariate methods estimate ranges directly. IF and GAD validate or challenge those ranges by flagging rows that look odd in context.

2. Compare everything at the row level first

The most useful common comparison unit is the row in your input table.

For each row, you will eventually have something like:

value	frequency	ageBand	gender	tukey_global	z_global	...	tukey
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From that, calculate summary indicators such as:

- global_consensus_score
- stratified_consensus_score
- ml_consensus_score Or just gad_if_agreement

This lets you ask:

- which values are globally implausible?
- which values are only implausible within strata?
- which values are flagged by context-aware ML but not by statistical thresholds?

That is the key comparison layer.

3. Compare methods by the ranges they imply

For the univariate methods, the first comparison is straightforward:

Create one table for the global analysis:

method	lowerBound	upperBound	width	excludedRows	excludedFrequency
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And one for the stratified analysis:

stratum	method	lowerBound	upperBound	width	excludedRows	excludedFrequency
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This lets you compare:

- which methods are narrowest or widest
- whether some methods consistently exclude more data
- whether age/gender materially changes the plausible range

That alone gives you one set of conclusions.

4. Use IF and GAD to assess contextual plausibility, not just to make another bound table

This is the important adjustment.

I would not ask IF and GAD to compete directly with Tukey or modified z-score on “who has the best min/max.” Instead, I would use them to answer:

Do the values retained by the univariate methods still look anomalous once age and gender context are considered?

That means IF and GAD help you evaluate whether a candidate range is too permissive or too restrictive.

For example:

- If a value is retained by most global univariate methods, but GAD and IF both flag it in a specific age/gender context, that suggests the global range may be too broad.

- If a value is excluded globally, but rarely flagged in the age/gender-aware models, that suggests the global method may be overcalling outliers.

So the ML models act like a contextual cross-check.

5. Build a comparison framework around four categories of values

This is where you start drawing conclusions.

For each row, classify it into one of these kinds of cases:

A. Consistently plausible

Not flagged globally, not flagged in strata, not flagged by IF/GAD.

These values define the core plausible range.

B. Globally implausible

Flagged by most global univariate methods and also usually flagged elsewhere.

These are your strongest candidates for truly implausible values.

C. Context-dependent

Not flagged globally, but flagged in age/gender strata and/or by IF/GAD.

These values may be plausible overall but implausible for specific demographic contexts.

D. Method-sensitive or borderline

Flagged by only one or two methods, or disagreement between global/statified/ML views.

These become your uncertainty zone.

That classification is much more informative than just comparing min/max tables.

6. Decide what final range you want to report

You probably need to be explicit that there are really two possible outputs:

A. Global plausible range

A single lower/upper bound for the measurement concept and unit.

This is easiest to report and useful for summary communication.

B. Contextual plausible ranges

Bounds that vary by age band and gender.

These are more biologically realistic, especially for systolic blood pressure.

I would recommend reporting both, with a clear interpretation:

- the **global range** as the overall summary
- the **stratified ranges** as the more clinically nuanced result

Then use IF and GAD to show whether contextual modeling supports the stratified view.

7. How to come to conclusions

This is the actual decision logic I would use.

Conclusion type 1: Does stratification matter?

Compare global versus stratified bounds.

If stratified bounds differ materially and in clinically sensible ways, then one conclusion is:

Age/gender stratification improves plausibility estimation for this measurement.

For systolic blood pressure, that would be a very plausible outcome.

Conclusion type 2: Which univariate methods are most stable?

Compare methods on:

- range width
- excluded frequency
- agreement with other methods
- stability across strata

If some methods produce very unstable or extreme bounds, you can conclude they are less suitable for this concept.

Conclusion type 3: Does contextual ML add information?

Look at rows retained by the univariate methods but flagged by IF/GAD.

If this happens often and in clinically interpretable ways, then you can conclude:

Context-aware models provide complementary information beyond value-only thresholds.

Conclusion type 4: What range should be recommended?

Your final recommended range should probably not come from a single method. It should come from a synthesis such as:

- use the **consensus of global univariate methods** for the broad concept-level range
- refine interpretation using **stratified univariate results**
- use **IF/GAD flags** to identify areas where global bounds appear too permissive or where contextual implausibility persists

In other words, your conclusion is likely to be a framework statement, not a winner-takes-all statement.

8. A practical reporting structure

Here is the simplest way to present results.

Table 1

Global univariate bounds by method

Table 2

Stratified univariate bounds by method and age/gender

Table 3

Agreement summary across methods

For example:

- percent of rows flagged by each method
- percent overlap with consensus
- percent flagged by IF and GAD among values retained by univariate methods

Figure 1

Distribution of values with global bounds overlaid

Figure 2

Stratified ranges by age band and gender

Figure 3

Heatmap or matrix of method agreement across rows

That will make your conclusions much easier to support.

9. My recommended decision rule

If you want one concrete way to synthesize all of this, I would use this logic:

1. Compute global univariate bounds for each method.
2. Compute stratified univariate bounds where sample size is sufficient.
3. For each row, derive:
 - global consensus flag
 - stratified consensus flag
 - IF flag
 - GAD flag
4. Label rows by combined evidence:
 - strong implausible: flagged by global consensus and stratified consensus, plus IF or GAD
 - likely implausible: flagged by one consensus layer and one ML method
 - contextual/borderline: flagged only in strata or only by ML
 - plausible: not flagged
5. Derive:
 - one global summary range
 - stratified ranges
6. Conclude whether context materially changes the plausible range and whether ML adds signal.

10. Bottom line

Yes, your plan is right.

To compare across everything you've done, do not ask "which method wins?" Instead ask:

- How do the **global and stratified ranges** differ?
- How much do methods **agree on flagged values**?
- Do IF and GAD identify **contextual anomalies not captured by univariate methods**?
- What **final plausible range or ranges** are most supported by the combined evidence?

That gives you a defensible conclusion such as:

Global univariate methods provide a useful first estimate of plausible bounds, but age/gender stratification yields more biologically realistic ranges, and IF/GAD provide additional contextual evidence for values that remain ambiguous under threshold-based methods.

If you want, I can turn this into a one-page analysis plan your team can review.



Please turn this into a one-page analysis plan with sample tables and figures.

Here is a **one-page analysis plan** you can share directly with your team. It's structured to be concise, actionable, and aligned with your study goal of learning plausible min/max values.

Analysis Plan: Learning Plausible Measurement Ranges

Objective

To estimate biologically plausible minimum and maximum values for a measurement concept (with a fixed unit) by comparing:

- **Global univariate outlier detection methods**
- **Stratified univariate methods (age × gender)**
- **Context-aware anomaly detection (Gaussian + Isolation Forest)**

and synthesizing results into a defensible plausible range.

Data

Input dataset (one row per value × context):

value	frequency	ageMidpoint	ageBand	gender
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Assumptions:

- Unit harmonized
- Frequency represents count of occurrences

Methods

1. Global Univariate Analysis

Apply each method to full dataset:

- Tukey
- Quantile thresholds
- Z-score
- Modified Z-score
- ESD

Output:

- Binary flag per row: `isOutlier_<method>_global`
- Derived bounds per method

2. Stratified Univariate Analysis

Repeat methods within each **ageBand × gender** stratum (if sufficient data).

Output:

- `isOutlier_<method>_strat`
- Bounds per stratum

3. Context-Aware Models

Gaussian Anomaly Detection (GAD)

- Inputs: value, ageMidpoint, gender
- Output: `isOutlier_gad`

Isolation Forest (IF)

- Inputs: value, frequency, ageMidpoint, gender
- Output: `isOutlier_if`

Harmonization Step

All methods produce:

Binary outlier flag per row

Construct:

- `global_consensus_score`
- `strat_consensus_score`
- `ml_flag = (gad OR if)`

Derived Classifications

Each row classified as:

Category	Definition
Plausible	Not flagged anywhere
Global Implausible	Flagged by global consensus
Contextual Implausible	Flagged only in stratified or ML
Strong Implausible	Flagged by global + strat + ML
Borderline	Partial disagreement

Range Estimation

Global Range

From rows where:

`global_consensus_score < threshold`

Copy

Stratified Range

From rows within each stratum where:

`strat_consensus_score < threshold`

Copy

Use:

- weighted quantiles (preferred)
- or min/max of retained values

 Sample Tables

Table 1 — Global Bounds

Method	Lower	Upper	Width	Excluded Freq
Tukey	90	210	120	45
Z-score	85	220	135	52
GAD	80	230	150	60
Consensus	88	215	127	48

Table 2 — Stratified Bounds

AgeBand	Gender	Method	Lower	Upper
18–39	M	Consensus	95	200
65–79	F	Consensus	100	220

Table 3 — Method Agreement

Metric	Value
% rows flagged (global consensus)	2.3%
% rows flagged (IF)	3.1%
% overlap (consensus $\cap$ IF)	1.8%

ChatGPT

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Table 4 — Value Classification

Value	Freq	AgeBand	Gender	Category
120	54	40–64	M	Plausible
260	1	65–79	F	Strong Implausible
145	2	18–39	F	Contextual

 Sample Figures

Figure 1 — Distribution + Global Bounds

- Histogram of values
- Overlay:
 - individual method bounds
 - consensus bounds

Figure 2 — Stratified Ranges

- Faceted plot by ageBand $\times$ gender
- Show consensus lower/upper bounds

Figure 3 — Method Agreement Heatmap

- Rows = values
- Columns = methods
- Color = outlier flag

Figure 4 — ML vs Consensus

- x-axis: value
- y-axis: IF score (or GAD score)
- Color: classification category

 Key Evaluation Questions

1. Do stratified bounds differ meaningfully from global bounds?
2. Which univariate methods are most stable across strata?
3. Do IF/GAD flag values retained by univariate methods?
4. Are flagged values clinically plausible or clearly erroneous?

 Expected Outcomes

- Consensus methods produce **stable plausible bounds**
 - Stratification yields **more biologically realistic ranges**
 - IF/GAD identify **additional contextual anomalies**
 - Combined approach improves distinction between:
 - implausible values
 - rare but valid values
-



Final Deliverables

- Global plausible range
 - Stratified plausible ranges
 - Method comparison summary
 - Classification of values
 - Recommendation on:
 - need for stratification
 - value of ML augmentation
-



Summary

This framework:

- Standardizes outputs across heterogeneous methods
 - Separates **global vs contextual plausibility**
 - Uses ML as a **complement, not replacement**
 - Produces interpretable and defensible ranges
-

If you want, I can next:

- Convert this into a **poster-ready Methods + Results layout**
- Or generate **R (HADES-style) scaffolding for the pipeline + tables/figures**

